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Professional Summary

Professional Engineer (P.E.) & Reliability Coordinator with 6 years of experience in Power System in Operation, Planning, Resource Integration, Modeling, Steady State & Dynamic Analysis at ERCOT ISO & LCRA TDSP. Master of Engineering from Lamar University Major in Power Electrical & Computer Engineering. Currently pursuing a Master of Science in Artificial Intelligence at University of Texas at Austin. AWS/IBM Certified.
Proficient in PSS/E, GE EMS SCADA/TSM/DTS, ABB MMS, Streamlit, Embeddings, Python (Pandas, NumPy, Matplotlib, Scikit-learn, XGBoost, Pytorch, Tensorflow Keras ,OpenAI, N8N), JavaScript, GitHub Pages.

Work History

Transmission Planning Model & Assessment Engineer | LCRA TSP | 3/2024 – Present

- Developed [AELAB](#) in Python automating Steady State Contingency Analysis, Dynamic Analysis, IDV generation, Contingency Generation & TPIT workflow.
- Review & approve Planned, For - Construction & Operational ratings for LCRA transmission lines & auto transformers & shunts resulting from substations, lines & auto transformers additions or upgrades.
- Prepare Transmission Project Information Tracking (TPIT) updates for internal & costumers' projects.
- Lead & present & assign tasks for planning team in multi department rating comparison meetings.
- Ensure system reliability, & compliance with NERC Standards, ERCOT Operation & Planning Guides.
- Maintain LCRA Planning Network Model in ERCOT according to capital projects in a timely manner.
- Participate in ERCOT SSWG, DWG, PLWG, LLWG, RPG, LFLTF working groups.
- Submit PMCR, DCP on ERCOT MOD for model changes & tuning.
- Propose & sponsor projects based on load forecast, generation & transmission capacity & budget.
- Perform Steady State Analysis for new Generation & Load Interconnect Requests.
- Perform Dynamic Stability Analysis for MOD-26, MOD-27, & Model Quality Test.
- Enhance model accuracy through data comparisons & validity checks.

Transmission Operation Network Model & EMS Engineer | LCRA TSP | 8/2022 – 3/2024

- Maintain LCRA Operation Network Model in ERCOT & LCRA EMS Model according to capital projects.
- Draft One Line Diagram for before & after network model changes for capital projects.
- Perform Contingency analysis for capital projects & outages & maintain State Estimator solutions.
- Submit Network Model Operation Requests (NOMCRs) & participate in ERCOT NDSWG working groups.
- Address real-time issues for SCADA, Transmission Security Management (TSM) applications & State Estimator.
- Maintain Dispatcher Training Simulator (DTS) system network model, data base & applications.
- Maintain PMU data in Epcdc & RTDMS server & client access manager.
- Update Line ratings & Impedances in EROT model & EMS based on Engineering team publications.
- Participate in network data working groups with ERCOT Collaborate with customers like PEC, BBEC, BEC, SBEC.

Real Time Power System Engineer | ERCOT ISO (CROSSTRaining) | 1/2022-4/2022

- Provide engineering support to ERCOT Control Room System Operators through Power Flow studies, Stability Assessments, & system applications support.
- Maintain Real-Time ERCOT State Estimator, Contingency Analysis, & Voltage/Transient Stability Analysis tools.

- Develop Constraint Management Plans such as TOAP based on engineering studies for grid vulnerabilities.
- Identify network model & applications quality issues.
- Collaborate with ERCOT System Operators & Market Participants to maintain grid reliability & security.
- Troubleshoot situational awareness tools & reported grid status & developments to ERCOT departments.

Operation Training Instructor | **ERCOT ISO** | **10/2020-8/2022**

- Develop power system simulation training scenarios to enhance ERCOT system operators' performance.
- Maintain EMS, MMS, & OTS systems, troubleshooted simulator issues.
- Prepare presentations for trainings & evaluate operator's responses during simulation trainings.
- Design simulations events for EEA, Black Start, RTA, IROL, Hurricane Drill, Low Inertia trainings.
- Participate as a RC, QSE or TO in real time simulations.
- Perform Contingency Analysis for DTS case preparation.

Power Electrical Engineer | **ERCOT ISO – SOAL technologies** | **10/2019 - 10/2020**

- Perform RARF registration & Reactive testing.
- Review & processed generation interconnection & full interconnection study (FIS) applications.
- Review QSA Full Interconnection Studies such as Short Circuit, Faciality, Steady State, Stability Studies.
- Utilize EMS & PSS/E Transmission Planning load flow cases for power system analysis.
- Perform Steady State N-1 & N-1-1 Contingency Analysis for Generation Interconnection Requests.

Associate Teacher | **HISD** | **2/2019 - 3/2022**

Substitute Teacher | **CFISD** | **4/2018-1/2019**

Substitute Teacher | **AISD** | **9/2025 - Present**

Education

M.S., Artificial Intelligence (GPA 4.0) | The University of Texas at Austin | 8/2024 – Present

Courses: Deep Learning, Machine Learning, Optimization, EAI, AIH, CSML,
Projects:

- Built a vision system & autonomous racing agent for SuperTuxKart, optimizing performance through advanced deep learning techniques.
- Applied machine learning algorithms to real-world data sets, solving problems in pattern recognition & dimensionality reduction.
- Developed ethical AI guidelines for system design, incorporating fairness & transparency into decision-making frameworks.

M.Eng., Electrical & Computer Engineering (GPA: 3.8) | Lamar University | 1/2019 - 5/2020

- Courses: Power System Motor & protection, Introduction to Robotics, Power Sys Stability & Control, Programmable Logic Controller, Computer Network I & II, Low Power CMOS Des & Rel, Cyber Physical Sys & Security, Instrumentation System & Auto.

B.S., in Electrical & Computer Engineering | Shahid Beheshti University | 10/2012 7/2017

- Courses: Protection & Relays, Power System I & II & labs, Electrical Machines I, II, III, Especial Machines & labs, Computer Architecture, Computer Programming, Linear Algebra, Electromagnetic, Industrial Drawing, System Analysis, Logical Circuits, Electronics 1 & 2, Telecommunications, Production & Power Station, High Pressure Plant Design & Project, Mathematics I, II & physics, Differential Equations, Statistics & Probability Engineering.

AI & Automation Projects

Technologies: Python, Streamlit, OpenAI API, Embeddings, PSS®E, NLP, Scikit-learn, XGBoost, HTML/CSS, JavaScript, GitHub Pages, Kaggle.

[Personal Portfolio Website](#) & [Resume & Portfolio Chatbot](#)

-Developed & deployed amirexirpe.com to showcase my resume, certifications, & AI-powered tools. Integrated a recruiter-facing chatbot trained on my experience & projects using semantic embeddings. The site includes interactive galleries, contact forms, downloadable documents, & iframe-embedded live apps.

[Hourly Load Forecast App \(AEP / PJM\)](#)

-Built a live load forecasting tool using PJM hourly data from Kaggle. Applied time-series feature engineering (lags, rolling averages, calendar variables) & trained an XGBoost model with low RMSE. Deployed with Streamlit & embedded into portfolio via iframe.

[ERCOT Grid Analytics & Forecasting Dashboard](#)

-Designed and deployed a real-time ERCOT data analytics platform using Streamlit, Plotly, and the ERCOT Public API to visualize system load, renewable generation, outages, and market prices.
-Implemented OAuth2.0 authentication (B2C flow) with token management and smart caching to handle ERCOT's API rate limits.
-Built an XGBoost-based load forecasting engine with time-series feature engineering (lags, rolling averages, cyclical encoding) and interactive hyperparameter tuning via sidebar controls.
-Added automatic train/test split validation with MAE, RMSE, and R^2 performance metrics, and dynamic Plotly diagnostics comparing actual vs forecasted loads.

[PSS®E Automation Assistant Bot](#), [PSS®E Multi Agent Automation Bot](#)

-Developed Copilot-style assistants that generate Python scripts for PSS®E tasks like contingency analysis, dynamic simulation, & model editing. Multi-agent version adds autonomous task by planning, retrieval & execution agents. Powered by the same end-to-end semantic search pipeline for high-precision technical retrieval.

[ERCOT Nodal Protocols, Planning Guides, DWG SSWG manuals & Resource Integration AI assistant](#)

-Built multiple GPT-powered assistants trained on ERCOT Planning Guides, Protocols, DWG/SSWG manuals, & interconnection processes. Used a custom embedding & retrieval pipeline to chunk, embed, & semantically search technical documents with OpenAI's text-embedding-3-small model & token-bounded cosine similarity. Supports compliance, model validation, & system integration analysis.

[Power Fault Classifier App](#)

-Created a Streamlit web app to classify power system faults using phasor measurements (Ia, Ib, Ic, Va, Vb, Vc). Trained & compared models (SVM, RF, MLP, XGBoost) with cross-validation & confusion matrix visualizations. Supports CSV uploads & result downloads.

[Power Grid GNN Alarm Prediction App](#)

Developed a GCN with focal loss and F1-optimized CV to predict voltage and thermal alarms on IEEE 14-bus scenarios. Implemented graph augmentation, dynamic topology mapping, and a linearized GCN toggle; deployed an interactive Streamlit dashboard with visual diagnostics and model artifact export.

[TinyLlama Fine-Tuning for Medical Q&A](#)

I fine-tuned the TinyLlama-1.1B-Chat model on 16K MedQuAD medical question-answer pairs using LoRA adapters, demonstrating that parameter-efficient methods can deliver measurable gains under compute constraints. By applying gradient checkpointing, 4-bit quantization, and an AdamW + warmup training schedule, I made training feasible on modest GPUs. The fine-tuned model achieved up to a 40% improvement in ROUGE-2 scores compared to the baseline and was published on the Hugging Face Hub ([tinyllama-medquad-lora](#)), where it has already been downloaded and used by external researchers. I also provided reproducible code, dataset splits, and documentation to support transparency and accessibility for the healthcare AI research community.

[Stock Market & Crypto AI Agent](#)

Developed an AI-powered trading assistant with Streamlit that connects to the Alpaca Paper Trading API for simulated equity trades, real-time market data, and SMA crossover strategy visualization. Integrated daily AI-generated market summaries automatically pushed to GitHub and [Telegram](#) via n8n workflows, enabling live content updates without manual intervention. Implemented adjustable moving-average parameters, demo/live mode toggling, and automated file versioning to support both public and private users in a cloud-deployed trading environment.

Licenses, Certifications, & skills

- P.E. License (Licensed Professional Engineer) – Texas Board of Professional Engineers #151267
- NERC System Operator Reliability Coordinator Certification- #RC 202105039
- Siemens PTI Academy Automating PSS®E Using Python (PSSC_625)
- AWS Certified Cloud Practitioner.
- Machine Learning with Python IBM Certification.
- Databases & SQL for Data Science with Python IBM Certification.
- Python for Data Science, AI & Development IBM Certification.
- Data Visualization with Python.
- Familiar with electrical standards & protocols (NEC NFPA, NERC, ERCOT, ANSI, IEEE).

Software

EMS GE Alstom, GE Reliance (PSLF, SOTE, TSM, DTSPSM, SCADA, RTNET/RTNA, STNET/STNA, RTCA, STCA)
MMS ABB (SCED, COP, RUC),
Python, MATLAB, SIMULINK, C++, Linux vi editor
PSS/E, PSLF, Power World, TARA, DmView, DWG True View, PI, Edna, Seeq, MMAP, Xmap & Gridgeo